

YaP MMO — multi-agent phase plan (M0-M4)

Goal: Ship a first-party, Folia-native **RuneScape-style progression stack** on YaPcore: skills, levels, XP curves, custom combat, crafting loops, quests/bosses — integrated with existing `yap-playerdata` , `yap-npcs` , `yap-tab` , YaPDB, and Link.

Not in scope for M0-M4: Bedrock skill UI parity, full RS quest count, PvP minigames (use future `yap-games`), factions (future `yap-factions`).

M5-M7 (extended): Bedrock MMO UI (M5), ability engine + 230+ combat abilities (M6), rich cast VFX + dedicated `CLAY_BALL` skill/ability icons (M7). See `docs/mmo/MMO_BEDROCK_UI.md` and `docs/mmo/MMO_ABILITIES.md` .

V1-V4 (ability VFX quality): Engine foundations → template overhaul → hero art → polish/crossplay. See `MMO_ABILITY_VFX.md` .

Global rules (every milestone)

- `folia-supported: true` in every new `plugin.yml`
- World/block/entity work via `com.yapcore.sched.YapSched` only
- DB via shared YaPDB (`use-shared-yapdb: true`) — same pattern as `playerdata-plugin`
- Register public APIs on Bukkit `ServicesManager` (`ServicePriority.Normal`)
- Gradle: add modules to `settings.gradle.kts` , wire `installProductDefaults` + `assembleRelease` when shipping jars
- Permissions: document in `docs/ops/PERMISSIONS.md`

Acceptance checklists under each milestone are **manual QA** for operator sign-off. Milestones M0-M7 are **shipped** (code + automated gates); tick boxes after live soak.

Reference implementations

Pattern	Copy from
API jar	<code>yap-first-party/api/yap-npcs-api/</code>
Plugin + shadow + Hikari	<code>yap-first-party/core-network/npcs-plugin/</code>
Job XP / level DB	<code>playerdata-plugin/.../JobRepository.java</code>
Quest packs	<code>npcs-plugin/.../quest/QuestPackLoader.java</code>
GUI menus	<code>playerdata-plugin/.../gui/Menus.java</code>
Dashboard snapshot	<code>DashboardEssentialsSnapshot.java</code>

Agent assignment map

Agent	Milestone	Theme	Depends on
Agent MMO-Kernel	M0	Skill engine + Mining end-to-end	Nothing
Agent MMO-Skills	M1	8 skills + /skills GUI + TAB	M0 merged
Agent MMO-Combat	M2	Custom combat, gear, food/pots	M0 merged; M1 combat XP hooks
Agent MMO-Craft	M3	Smithing/cooking/crafting + economy	M0 + M1
Agent MMO-Content	M4	Quests v2, bosses, areas, hiscores	M0-M3

Parallelism: M2 and M3 can start once **M0 is merged** if they stub against `SkillService` / `CombatService` interfaces from M0. Full integration tests need M1.

Module layout (created incrementally)

```
yap-first-party/api/yap-mmo-api/           # M0 – interfaces + records
yap-first-party/api/yap-abilities-api/      # M6 – ability + status effect API
yap-first-party/api/yap-bedrock-ui-api/     # M5 – Bedrock form/action bar API
yap-first-party/gameplay/skills-plugin/     # M0-M1 – yap-skills.jar
yap-first-party/gameplay/combat-plugin/     # M2 – yap-combat.jar
yap-first-party/gameplay/crafting-plugin/   # M3 – yap-crafting.jar
yap-first-party/gameplay/mmo-content-plugin/ # M4 – yap-mmo-content.jar
yap-first-party/gameplay/abilities-plugin/  # M6-M7 – yap-abilities.jar
yap-first-party/gameplay/mmo-bedrock-plugin/ # M5 – yap-mmo-bedrock.jar
yap-first-party/core-network/bedrock-ui-plugin/ # M5 – yap-bedrock-ui.jar
```

Gradle project names (suggested):

- `:yap-mmo-api` → `yap-mmo-api.jar`
- `:skills-plugin` → `yap-skills.jar`
- `:combat-plugin` → `yap-combat.jar`
- `:crafting-plugin` → `yap-crafting.jar`
- `:mmo-content-plugin` → `yap-mmo-content.jar`

Ship M0-M1 under `installGameplayDefaults` (opt-in gameplay). M2-M4 same bucket.

M0 — Skill engine kernel (one skill: Mining)

Status:  Shipped (M0)

Objective

Prove the full loop: **config skill** → **action** → **XP** → **level-up** → **DB persist** → **GUI read** → **quest reward hook** using **Mining only**.

Deliverables

1. yap-mmo-api

```
// Core types (package com.yapcore.mmo)
SkillId, SkillDefinition, XpTable, SkillProgress, LevelUpEvent
SkillService {
    CompletableFuture<SkillProgress> get(UUID, SkillId);
    CompletableFuture<Void> addXp(UUID, SkillId, double amount, XpSource source);
    int levelForXp(SkillId, double xp);
    double xpForLevel(SkillId, int level);
    Collection<SkillDefinition> definitions();
}
XpSource { action, quest, admin, command }
```

- XP table: RS-style exponential curve in YAML (base-xp , growth , max-level: 99)
- Event: SkillLevelUpEvent (Bukkit) for TAB/combat plugins

2. yap-skills plugin

Area	Detail
DB	Table yap_skill_progress (uuid, skill_id, xp, level, updated_at)
Config	skills/mining.yml — break map Material → xp , optional level gates
Listener	BlockBreakEvent (MONITOR) — ore types grant mining XP; region-safe
Commands	/skills [player] , /skill addxp <player> <skill> <amount> , /yskills reload
GUI	Chest menu: one row per skill (M0: only Mining slot live)
Feedback	Action bar on XP drip; title/chat on level-up
PAPI	%yapskill_<id>_level% , %yapskill_<id>_xp% , %yapskill_total_level% (total=sum)
Services	Register SkillService on enable

3. Integration stub (M0 minimal)

- **npcs:** Add quest reward token `skill_xp:mining:500` in `QuestServiceImpl.dispatchRewards` (compileOnly `yap-mmo-api` ; no-op if skills jar absent)
- **playerdata:** Document that legacy `jobs.miner` stays off when skills enabled; do not merge job XP yet

4. Smoke + tests

- Unit: `XpTableTest` — level 1, 50, 99 boundaries
- Unit: `MiningXpCalculatorTest` — material map

Acceptance

- [] Break iron ore → mining XP increases in DB within 2s
- [] Level-up fires `SkillLevelUpEvent` and shows player feedback
- [] `/skills` shows Mining level + XP bar
- [] `/skill addxp` works with `yapskills.admin`
- [] Reload does not duplicate listeners
- [] All block checks use `YapSched` ; DB on async executor
- [] `gradle :skills-plugin:installIntoPlugins` + gameplay install puts jar in `plugins/`

Agent prompt (copy-paste)

```
Implement YaP MMO Phase M0 per docs/mmo/MMO_PHASES.md.
Create yap-mmo-api + skills-plugin (Mining only). Folia-safe YapSched.
DB table yap_skill_progress, SkillService on ServicesManager, /skills GUI,
BlockBreakEvent XP, PAPI placeholders, `gradle installGameplayDefaults`.
Extend yap-npcs quest rewards with skill_xp:mining:N (soft dependency).
Do not implement other skills yet. Match npcs-plugin Gradle/shadow patterns.
```

Key files to create

- `yap-first-party/api/yap-mmo-api/src/main/java/com/yapcore/mmo/SkillService.java`
- `yap-first-party/gameplay/skills-plugin/src/main/java/com/yapcore/skills/SkillsPlugin.java`
- `yap-first-party/gameplay/skills-plugin/src/main/resources/skills/mining.yml`

M1 — Eight skills + progression UI

Status:  Shipped (M1)

Agent: MMO-Skills

Depends on: M0 merged

Ships: Updated `yap-skills.jar` , finetune module optional

Objective

Expand kernel to **8 RuneScape-style skills** with shared XP tables, requirements, and network-visible progression (TAB + dashboard read-only).

Skills (v1 set)

Skill	XP sources
Mining	(M0) ore break
Woodcutting	log break
Fishing	fish catch (<code>PlayerFishEvent</code>)
Cooking	smelt/craft cooking recipes (simple: cook raw food in furnace click or custom)
Smithing	smelt bars + basic smith actions (prep for M3; grant XP on bar smelt only in M1)
Attack	mob damage dealt (stub multiplier until M2 combat)
Strength	mob damage dealt (split XP 50/50 with Attack for now)
Defence	damage taken from mobs
Hitpoints	1/3 combat XP to HP (RS-style ratio placeholder)

Use **config-driven** skill defs: `skills/*.yaml` one file per skill.

Deliverables

Area	Detail
Listeners	Woodcut, Fish, FurnaceExtract/Smelt (cooking/smithing), EntityDamageByEntity, EntityDamageEvent
Requirements	<code>SkillRequirement</code> — <code>requires: { mining: 15 }</code> on skill actions (config)
Commands	<code>/skill top <skill> [page]</code> , <code>/skill set <player> <skill> <level></code>
GUI	Full <code>/skills</code> — 8 icons, progress bars (item meta or Adventure boss bar in menu)
TAB	Listen <code>SkillLevelUpEvent</code> → refresh sidebar; placeholders for combat level
Combat level	Formula in config: <code>floor((attack+strength+defence+hitpoints)/4)</code> display only
Dashboard	GET <code>/api/mmo</code> snapshot: skill count, online sample levels (read-only)
Docs	<code>docs/mmo/MMO_SKILLS.md</code> — skill list, XP sources, permissions

Acceptance

- [] Each skill gains XP from its documented action at least once (manual test checklist in doc)
- [] Level requirements block action with clear message (e.g. mine mithril needs 55 mining — use config test ore)
- [] `/skill top mining` returns DB leaderboard (paginated, async)
- [] TAB sidebar updates on level-up
- [] Dashboard `/api/mmo` returns JSON without server crash when skills plugin loaded

Agent prompt

```
Implement YaP MMO Phase M1 per docs/mmo/MMO_PHASES.md on top of merged M0.
Add 7 more skills (woodcutting, fishing, cooking, smithing smelt XP, attack, strength,
defence, hitpoints). Config-driven skills/*.yml, /skill top, full /skills GUI,
combat level display, TAB refresh on SkillLevelUpEvent, dashboard GET /api/mmo.
Do not implement custom damage formulas (M2). Folia-safe throughout.
```

M2 — Custom combat system

Status:  Shipped (M2)

Agent: MMO-Combat

Depends on: **M0 merged; M1** for combat skill XP hooks

Ships: `yap-combat.jar`

Docs: [MMO_COMBAT.md](#)

Objective

Replace vanilla mob PvE math with **configurable combat**: accuracy, max hit, defence, gear bonuses, food, potions. Feed Attack/Strength/Defence/Hitpoints XP from M1.

Deliverables

`yap-mmo-api` extensions

```
CombatService {
    CombatStats stats(Player);
    void recalculate(Player); // on equip, buff expiry
}
CombatStats { attack, strength, defence, hitpoints, prayer, ranged, magic } //
prayer/ranged/magic stubs 1-99 from gear only
GearBonus { attackBonus, strengthBonus, defenceBonus, ... } // aggregated from item NBT or
config material map
```

yap-combat plugin

Area	Detail
Hit pipeline	<code>EntityDamageByEntityEvent</code> — cancel vanilla, apply formula damage on entity thread
Formula	Config: <code>maxHit = floor((strength + gear) * levelFactor)</code> ; hit roll vs defence
Gear	<code>items.yml</code> — material → bonuses; optional PDC key <code>yap_gear_tier</code> for custom items
Food	Right-click food → heal HP (custom HP bar separate from vanilla 20 hearts OR scale 10 HP = 1 heart) — pick one in config
Potions	2-3 boost types (attack pot, strength pot, defence pot) — duration + cooldown in config
Death	RS-lite: keep inventory config flag; respawn at spawn; HP restore
PvP	Config off by default; claim flag <code>pvp-combat: allow</code> hooks into <code>yap-playerdata</code> claims
XP	On kill: split XP to combat skills via <code>SkillService.addXp</code>
Commands	<code>/combat stats</code> , <code>/combat reload</code> , <code>/yapcombat admin sethp <player> <hp></code>

Integration

- **guard-plugin:** Optional — ignore players with combat invuln admin flag
- **npcs:** Boss entities tagged via PDC `yap_boss_id`

Acceptance

- [] Zombie fight uses custom damage; vanilla sword spam DPS changes with strength level
- [] Equipping iron → diamond armor changes defence and reduces hits taken
- [] Food heals custom HP; cannot spam (tick delay)
- [] Potion buff visible in `/combat stats`
- [] Combat XP awards on mob kill match M1 ratios
- [] Folia: no cross-region entity mutation without `YapSched.entity`

Agent prompt

Implement YaP MMO Phase M2 per docs/mmo/MMO_PHASES.md.
Create yap-combat.jar + CombatService API. Custom PvE damage pipeline, gear bonuses from config, food + 3 potions, custom HP model (document choice), combat XP via SkillService. PvP off default. Folia-safe YapSched. Unit tests for damage formula.

M3 — Crafting & economy loops

Status:  Shipped (M3)

Agent: MMO-Craft

Depends on: M0 + M1

Ships: yap-crafting.jar

Docs: crafting recipes in `plugins/YaPCrafting/recipes/`

Objective

Close the **gather** → **process** → **equip** loop: smithing bars → items, cooking fish, recipe unlocks by level, shop/AH hooks.

Deliverables

Area	Detail
Recipe engine	YAML recipes: <code>id</code> , <code>type: SMITHING COOKING CRAFTING</code> , <code>level</code> , <code>inputs[]</code> , <code>output</code> , <code>xp</code>
Stations	Anvil / furnace / crafting table interact — match recipe by ingredients + skill level
Smithing	Full bar → tool/armor tier progression (config tiers aligned with combat gear in M2)
Cooking	Raw fish → cooked; burn chance below level threshold
Unlocks	<code>/recipe list <skill></code> , action bar on discover
Economy	Sell prices in config; optional <code>/sell</code> hand-in (uses playerdata <code>BalanceStore</code> via API)
NPC traders	Extend <code>yap-playerdata</code> traders OR crafting shop GUI for test NPC
AH	List crafted gear on playerdata auctions (if <code>features.auctions: true</code>) — document only if not wired

Acceptance

- [] Mine ore (mining) → smelt bar (smithing XP) → smith dagger (smithing XP + item with gear stats for M2)
- [] Fish → cook → eat (fishing + cooking XP loop)
- [] Recipe blocked below required level with message citing required level
- [] `/sell` adds money when economy enabled

Agent prompt

```
Implement YaP MMO Phase M3 per docs/mmo/MMO_PHASES.md.  
Create yap-crafting.jar with YAML recipe engine (smithing, cooking, crafting), station
```


listeners, level gates, /recipe list, /sell via playerdata BalanceStore.
Integrate gear output with yap-combat items.yml tiers. Folia-safe.

M4 — Content layer (quests, bosses, areas, hiscores)

Status:  Shipped (M4)

Objective

Ship **playable content** on top of systems: extended quests, 3 boss fights, 2 skill areas, global hiscores, optional world markers.

Deliverables

Quest system v2 (extend yap-npcs)

New objective types in quest packs:

Type	Example
SKILL_LEVEL	mining >= 20
CRAFT_ITEM	craft iron_dagger
KILL_BOSS	kill boss goblin_king
GATHER	(existing break/kill)

Rewards: skill_xp, item, money, unlock_recipe, teleport_unlock

Boss module (in content plugin or combat)

- 3 bosses in YAML: stats, spawn location, loot table, respawn timer
- PDC tag yap_boss_id; combat plugin damage rules apply
- Loot: custom gear pieces for M2 tiers

Skill areas

- **Mining guild zone** — flag region + ore respawn nodes (regenerating ore facade: stone → ore on timer)
- **Fishing spot** — higher XP rate in tagged water region (RegionInteract from yap-regions or cuboid in config)

Hiscores

- `/hiscores <skill>` — top 50 global (DB query async)
- Link broadcast optional: level 99 announcements
- Dashboard tab **MMO** (extend M1): hiscore preview, boss kill counts

Content pack

Ship `plugins/yap-mmo-content/quests/starter_chain.yml` — 5-quest chain teaching loop

Acceptance

- [] Complete starter quest chain grants ≥ 500 XP across 2 skills + one recipe unlock
- [] One boss killable with M2 combat; drops usable gear
- [] Mining guild ore respawns after configured delay
- [] `/hiscores mining` works cross-restart

Agent prompt

Implement YaP MMO Phase M4 per docs/mmo/MMO_PHASES.md.
Create yap-mmo-content.jar: quest v2 objectives (extend yap-npcs quest packs), 3 bosses, 2 skill areas, `/hiscores`, `starter_chain.yml` content pack, dashboard MMO tab hiscore preview. Depends on yap-skills, yap-combat, yap-crafting. Folia-safe.

M5 — Bedrock MMO UI

Status:  Shipped (M5)

Ships: `yap-bedrock-ui.jar`, `yap-mmo-bedrock.jar`

Docs: [MMO_BEDROCK_UI.md](#)

FormService skills hub, paginated recipes/hiscores, XP action bar mirror, combat sidebar.

M6 — Ability engine (230+ combat abilities)

Status:  Shipped (M6)

Ships: `yap-abilities-api.jar`, `yap-abilities.jar`

Docs: [MMO_ABILITIES.md](#)

YAML-driven abilities: projectiles, VFX, buffs/debuffs, cooldowns, `/ability`, `/cast` delegation.

M7 — Advanced mechanics + client graphics

Status:  Shipped (M7)

Ships: updated `yap-abilities.jar`, `yap-mmo-bedrock.jar`, `resourcepacks/yap-abilities/overlay`

AoE, homing projectiles, chain lightning, cast conditions, animation sync, blaze-rod spell icons, Bedrock spellbook panel. Smoke: gradle installGameplayDefaults

V1-V4 — Ability VFX quality

Status:  Complete

Docs: [MMO_ABILITY_VFX.md](#)

Ships: updated `yap-abilities-api.jar`, `yap-abilities.jar`, regenerated bulk packs, `showcase_vfx.yml`, `showcase_heroes.yml`, hero icons, soak gates

Raises cast/travel/hit visuals from generated templates to top-MMO-plugin quality in four phases (engine → kits → heroes → polish).

```
python3 scripts/generate-ability-pack.py
python3 scripts/generate-hero-ability-icons.py
python3 scripts/content/ability-vfx-soak-gate.py
```

Cross-milestone checklist (release manager)

After **M7**, run full integration:

Update:

- [YAPCORE_WHITEPAPER.md](#) — MMO section
 - `docs/ops/PERMISSIONS.md` — `yapskills.*`, `yapcombat.*`, `yapcraft.*`, `yapmmo.*`, `yapabilities.*`
 - `gradle/yap-product.gradle.kts` — gameplay install includes all MMO jars
 - `gradle/yap-release.gradle.kts` — release artifact list
 - `docs/mmo/MMO_PHASES.md` — milestone status
-

Dependency graph

```
flowchart LR
    M0[M0 Kernel]
    M1[M1 Skills]
    M2[M2 Combat]
    M3[M3 Crafting]
    M4[M4 Content]
    M5[M5 Bedrock UI]
    M6[M6 Abilities]
    M7[M7 Mechanics]
    M0 --> M1
    M0 --> M2
    M1 --> M2
    M0 --> M3
    M1 --> M3
    M2 --> M4
    M3 --> M4
    M1 --> M4
    M0 --> M4
    M1 --> M5
    M2 --> M5
    M0 --> M6
    M2 --> M6
    M6 --> M7
    M5 --> M7
```

Config flags (network operator)

```
# yap-skills/config.yml
enabled: true
max-level: 99
skills:
  mining:
    enabled: true
# yap-combat/config.yml
custom-hp:
  enabled: true
  hearts-display: 10 # 10 custom HP per vanilla heart
pvp: false
# yap-crafting/config.yml
recipes-directory: recipes
```

Coexistence: When yap-skills enabled, set playerdata features.jobs: false to avoid double-paying miners.

Estimated effort (planning only)

Milestone	Calendar (focused agent)
M0	3-7 days
M1	7-14 days
M2	7-14 days
M3	7-10 days
M4	10-14 days

Total: ~6-10 weeks sequential; **M2 || M3** after M1 saves ~1-2 weeks.